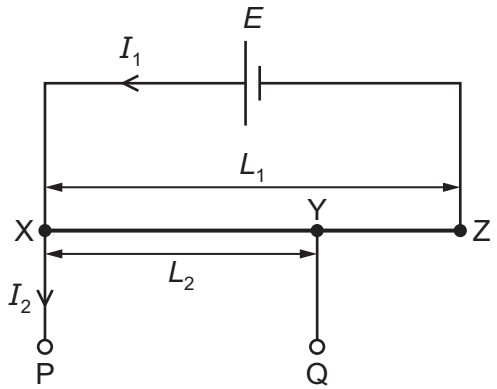
**35** The diagram shows part of a circuit that uses a potentiometer wire to measure a potential difference (p.d.) in an external circuit.




The potentiometer wire XZ has length $L_1$. It is connected to a cell of known electromotive force (e.m.f.) $E$ that has negligible internal resistance.

Terminals P and Q are connected to the external p.d. to be measured. The length of the potentiometer wire between points X and Y is $L_2$.

The ratio of the lengths $L_1$ and $L_2$ is used to determine the p.d. between P and Q in terms of $E$.

Which condition must be met in order to determine this p.d.?

**A** The current $I_1$ must be zero.

**B** The current $I_2$ must be zero.

**C** The p.d. across YZ must be zero.

**D** The resistance of the external circuit must be zero.